

IIT JAM 2017 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS

60

TOTAL MARKS

100 M

TEST DURATION

180 Min

PAPER PATTERN

MCQ • MSQ • NAT
(60 Ques)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2017 • Q1 • ID: JAM_2017_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 Consider the function $f(x, y) = 5 - 4 \sin x + y^2$ for $0 < x < 2\pi$ and $y \in \mathbb{R}$. The set of critical points of $f(x, y)$ consists of

- (A) a point of local maximum and a point of local minimum
- (B) a point of local maximum and a saddle point
- (C) a point of local maximum, a point of local minimum and a saddle point
- (D) a point of local minimum and a saddle point

Question 2 (JAM 2017 • Q2 • ID: JAM_2017_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that φ' is strictly increasing with $\varphi'(1) = 0$. Let α and β denote the minimum and maximum values of $\varphi(x)$ on the interval $[2, 3]$, respectively. Then which one of the following is TRUE?

- (A) $\beta = \varphi(3)$ (B) $\alpha = \varphi(2.5)$ (C) $\beta = \varphi(2.5)$ (D) $\alpha = \varphi(3)$

Question 3 (JAM 2017 • Q3 • ID: JAM_2017_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 The number of generators of the additive group $\mathbb{Z}_{36}$ is equal to

- (A) 6 (B) 12 (C) 18 (D) 36

Question 4 (JAM 2017 • Q4 • ID: JAM_2017_Q4)

MCQ

+1 M

Neg: -0.33

Q.4

$$\lim_{n \rightarrow \infty} \frac{\pi}{n} \sum_{k=1}^n \sin\left(\frac{\pi}{2} + \frac{5\pi}{2} \cdot \frac{k}{n}\right) =$$

- (A) $\frac{2\pi}{5}$ (B) $\frac{5}{2}$ (C) $\frac{2}{5}$ (D) $\frac{5\pi}{2}$

Question 5 (JAM 2017 • Q5 • ID: JAM_2017_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $g(u, v) = f(u^2 - v^2)$, then

$$\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} =$$

- (A) $4(u^2 - v^2)f''(u^2 - v^2)$
- (B) $4(u^2 + v^2)f''(u^2 - v^2)$
- (C) $2f'(u^2 - v^2) + 4(u^2 - v^2)f''(u^2 - v^2)$
- (D) $2(u - v)^2 f''(u^2 - v^2)$

Question 6 (JAM 2017 • Q6 • ID: JAM_2017_Q6)

MCQ

+1 M

Neg: -0.33

Q.6

$$\int_0^1 \int_x^1 \sin(y^2) dy dx =$$

- (A) $\frac{1+\cos 1}{2}$
- (B) $1 - \cos 1$
- (C) $1 + \cos 1$
- (D) $\frac{1-\cos 1}{2}$

Question 7 (JAM 2017 • Q7 • ID: JAM_2017_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 Let $f_1(x), f_2(x), g_1(x), g_2(x)$ be differentiable functions on $\mathbb{R}$. Let $F(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$ be the determinant of the matrix $\begin{bmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{bmatrix}$. Then $F'(x)$ is equal to

- (A) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2'(x) & g_2(x) \end{vmatrix}$
- (B) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$
- (C) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} - \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$
- (D) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$

Question 8 (JAM 2017 • Q8 • ID: JAM_2017_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 Let

$$f(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right), \quad x \neq 0.$$

Write $L = \lim_{x \rightarrow 0^-} f(x)$ and $R = \lim_{x \rightarrow 0^+} f(x)$. Then which one of the following is TRUE?

- (A) L exists but R does not exist
- (B) L does not exist but R exists
- (C) Both L and R exist
- (D) Neither L nor R exists

Question 9 (JAM 2017 • Q9 • ID: JAM_2017_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 If $\lim_{T \rightarrow \infty} \int_0^T e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, then

$$\lim_{T \rightarrow \infty} \int_0^T x^2 e^{-x^2} dx =$$

- (A) $\frac{\sqrt{\pi}}{4}$ (B) $\frac{\sqrt{\pi}}{2}$ (C) $\sqrt{2\pi}$ (D) $2\sqrt{\pi}$

Question 10 (JAM 2017 • Q10 • ID: JAM_2017_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 If

$$f(x) = \begin{cases} 1+x & \text{if } x < 0 \\ (1-x)(px+q) & \text{if } x \geq 0 \end{cases}$$

satisfies the assumptions of Rolle's theorem in the interval $[-1, 1]$, then the ordered pair (p, q) is

- (A) $(2, -1)$ (B) $(-2, -1)$ (C) $(-2, 1)$ (D) $(2, 1)$

Question 11 (JAM 2017 • Q11 • ID: JAM_2017_Q11)

MCQ

+2 M

Neg: -0.67

Q.11 The flux of the vector field

$$\vec{F} = \left(2\pi x + \frac{2x^2 y^2}{\pi} \right) \hat{i} + \left(2\pi xy - \frac{4y}{\pi} \right) \hat{j}$$

along the outward normal, across the ellipse $x^2 + 16y^2 = 4$ is equal to

- (A) $4\pi^2 - 2$ (B) $2\pi^2 - 4$ (C) $\pi^2 - 2$ (D) 2π

Question 12 (JAM 2017 • Q12 • ID: JAM_2017_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let $\mathcal{M}$ be the set of all invertible 5×5 matrices with entries 0 and 1. For each $M \in \mathcal{M}$, let $n_1(M)$ and $n_0(M)$ denote the number of 1's and 0's in M , respectively. Then

$$\min_{M \in \mathcal{M}} |n_1(M) - n_0(M)| =$$

- (A) 1 (B) 3 (C) 5 (D) 15

Question 13 (JAM 2017 • Q13 • ID: JAM_2017_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let $M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 2 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then

$$\lim_{n \rightarrow \infty} M^n x$$

(A) does not exist

(B) is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (C) is $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ (D) is $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Question 14 (JAM 2017 • Q14 • ID: JAM_2017_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 Let $\vec{F} = (3 + 2xy)\hat{i} + (x^2 - 3y^2)\hat{j}$ and let L be the curve

$$\vec{r}(t) = e^t \sin t \hat{i} + e^t \cos t \hat{j}, \quad 0 \leq t \leq \pi.$$

Then

$$\int_L \vec{F} \cdot d\vec{r} =$$

- (A) $e^{-3\pi} + 1$ (B) $e^{-6\pi} + 2$ (C) $e^{6\pi} + 2$ (D) $e^{3\pi} + 1$

Question 15 (JAM 2017 • Q15 • ID: JAM_2017_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 The line integral of the vector field

$$\vec{F} = zx \hat{i} + xy \hat{j} + yz \hat{k}$$

along the boundary of the triangle with vertices $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$, oriented anti-clockwise, when viewed from the point $(2,2,2)$, is

- (A) $\frac{-1}{2}$ (B) -2 (C) $\frac{1}{2}$ (D) 2

Question 16 (JAM 2017 • Q16 • ID: JAM_2017_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 The area of the surface $z = \frac{xy}{3}$ intercepted by the cylinder $x^2 + y^2 \leq 16$ lies in the interval

- (A) $(20\pi, 22\pi]$ (B) $(22\pi, 24\pi]$ (C) $(24\pi, 26\pi]$ (D) $(26\pi, 28\pi]$

Question 17 (JAM 2017 • Q17 • ID: JAM_2017_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 For $a > 0, b > 0$, let $\vec{F} = \frac{x\hat{j} - y\hat{i}}{b^2x^2 + a^2y^2}$ be a planar vector field. Let

$$C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = a^2 + b^2\}$$

be the circle oriented anti-clockwise. Then

$$\oint_C \vec{F} \cdot d\vec{r} =$$

(A) $\frac{2\pi}{ab}$

(B) 2π

(C) $2\pi ab$

(D) 0

Question 18 (JAM 2017 • Q18 • ID: JAM_2017_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 The flux of $\vec{F} = y\hat{i} - x\hat{j} + z^2\hat{k}$ along the outward normal, across the surface of the solid

$$\{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq \sqrt{2 - x^2 - y^2}\}$$

is equal to

(A) $\frac{2}{3}$

(B) $\frac{5}{3}$

(C) $\frac{8}{3}$

(D) $\frac{4}{3}$

Question 19 (JAM 2017 • Q19 • ID: JAM_2017_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(2) = 2$ and

$$|f(x) - f(y)| \leq 5(|x - y|)^{3/2}$$

for all $x \in \mathbb{R}, y \in \mathbb{R}$. Let $g(x) = x^3 f(x)$. Then $g'(2) =$

(A) 5

(B) $\frac{15}{2}$

(C) 12

(D) 24

Question 20 (JAM 2017 • Q20 • ID: JAM_2017_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let $f: \mathbb{R} \rightarrow [0, \infty)$ be a continuous function. Then which one of the following is NOT TRUE?

- (A) There exists $x \in \mathbb{R}$ such that $f(x) = \frac{f(0)+f(1)}{2}$
- (B) There exists $x \in \mathbb{R}$ such that $f(x) = \sqrt{f(-1)f(1)}$
- (C) There exists $x \in \mathbb{R}$ such that $f(x) = \int_{-1}^1 f(t)dt$
- (D) There exists $x \in \mathbb{R}$ such that $f(x) = \int_0^1 f(t)dt$

Question 21 (JAM 2017 • Q21 • ID: JAM_2017_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 The interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{1}{(-3)^{n+2}} \frac{(4x-12)^n}{n^2+1}$$

is

- (A) $\frac{10}{4} \leq x < \frac{14}{4}$
- (B) $\frac{9}{4} \leq x < \frac{15}{4}$
- (C) $\frac{10}{4} \leq x \leq \frac{14}{4}$
- (D) $\frac{9}{4} \leq x \leq \frac{15}{4}$

Question 22 (JAM 2017 • Q22 • ID: JAM_2017_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $\mathcal{P}_3$ denote the real vector space of all polynomials with real coefficients of degree at most 3. Consider the map $T: \mathcal{P}_3 \rightarrow \mathcal{P}_3$ given by $T(p(x)) = p''(x) + p(x)$. Then

- (A) T is neither one-one nor onto
- (B) T is both one-one and onto
- (C) T is one-one but not onto
- (D) T is onto but not one-one

Question 23 (JAM 2017 • Q23 • ID: JAM_2017_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $f(x, y) = \frac{x^2 y}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Then

- (A) $\frac{\partial f}{\partial x}$ and f are bounded
- (B) $\frac{\partial f}{\partial x}$ is bounded and f is unbounded
- (C) $\frac{\partial f}{\partial x}$ is unbounded and f is bounded
- (D) $\frac{\partial f}{\partial x}$ and f are unbounded

Question 24 (JAM 2017 • Q24 • ID: JAM_2017_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 Let S be an infinite subset of $\mathbb{R}$ such that $S \setminus \{\alpha\}$ is compact for some $\alpha \in S$. Then which one of the following is TRUE?

- (A) S is a connected set
- (B) S contains no limit points
- (C) S is a union of open intervals
- (D) Every sequence in S has a subsequence converging to an element in S

Question 25 (JAM 2017 • Q25 • ID: JAM_2017_Q25)

MCQ

+2 M

Neg: -0.67

Q.25

$$\sum_{n=1}^{\infty} \tan^{-1} \frac{2}{n^2} =$$

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{3\pi}{4}$

(D) π

Question 26 (JAM 2017 • Q26 • ID: JAM_2017_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let $0 < a_1 < b_1$. For $n \geq 1$, define

$$a_{n+1} = \sqrt{a_n b_n} \quad \text{and} \quad b_{n+1} = \frac{a_n + b_n}{2}.$$

Then which one of the following is NOT TRUE?

- (A) Both $\{a_n\}$ and $\{b_n\}$ converge, but the limits are not equal
- (B) Both $\{a_n\}$ and $\{b_n\}$ converge and the limits are equal
- (C) $\{b_n\}$ is a decreasing sequence
- (D) $\{a_n\}$ is an increasing sequence

Question 27 (JAM 2017 • Q27 • ID: JAM_2017_Q27)

MCQ

+2 M

Neg: -0.67

Q.27

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(\frac{1}{\sqrt{3} + \sqrt{6}} + \frac{1}{\sqrt{6} + \sqrt{9}} + \cdots + \frac{1}{\sqrt{3n} + \sqrt{3n+3}} \right) =$$

(A) $1 + \sqrt{3}$

(B) $\sqrt{3}$

(C) $\frac{1}{\sqrt{3}}$

(D) $\frac{1}{1+\sqrt{3}}$

Question 28 (JAM 2017 • Q28 • ID: JAM_2017_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 Which one of the following is TRUE?

(A) Every sequence that has a convergent subsequence is a Cauchy sequence

(B) Every sequence that has a convergent subsequence is a bounded sequence

(C) The sequence $\{\sin n\}$ has a convergent subsequence(D) The sequence $\left\{n \cos \frac{1}{n}\right\}$ has a convergent subsequence

Question 29 (JAM 2017 • Q29 • ID: JAM_2017_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 A particular integral of the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^{2x} \sin x$$

is

- (A) $\frac{e^{2x}}{10} (3 \cos x - 2 \sin x)$
- (B) $-\frac{e^{2x}}{10} (3 \cos x - 2 \sin x)$
- (C) $-\frac{e^{2x}}{5} (2 \cos x + \sin x)$
- (D) $\frac{e^{2x}}{5} (2 \cos x - \sin x)$

Question 30 (JAM 2017 • Q30 • ID: JAM_2017_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let $y(x)$ be the solution of the differential equation

$$(xy + y + e^{-x})dx + (x + e^{-x})dy = 0$$

satisfying $y(0) = 1$. Then $y(-1)$ is equal to

- (A) $\frac{e}{e-1}$ (B) $\frac{2e}{e-1}$ (C) $\frac{e}{1-e}$ (D) 0

Question 31 (JAM 2017 • Q31 • ID: JAM_2017_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31 For $\alpha, \beta \in \mathbb{R}$, define the map $\varphi_{\alpha, \beta}: \mathbb{R} \rightarrow \mathbb{R}$ by $\varphi_{\alpha, \beta}(x) = \alpha x + \beta$. Let

$$G = \{\varphi_{\alpha, \beta} \mid (\alpha, \beta) \in \mathbb{R}^2\}$$

For $f, g \in G$, define $g \circ f \in G$ by $(g \circ f)(x) = g(f(x))$. Then which of the following statements is/are TRUE?

- (A) The binary operation $\circ$ is associative
- (B) The binary operation $\circ$ is commutative
- (C) For every $(\alpha, \beta) \in \mathbb{R}^2$, $\alpha \neq 0$ there exists $(a, b) \in \mathbb{R}^2$ such that $\varphi_{\alpha, \beta} \circ \varphi_{a, b} = \varphi_{1, 0}$
- (D) $(G, \circ)$ is a group

Question 32 (JAM 2017 • Q32 • ID: JAM_2017_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 The volume of the solid

$$\left\{ (x, y, z) \in \mathbb{R}^3 \mid 1 \leq x \leq 2, \quad 0 \leq y \leq \frac{2}{x}, \quad 0 \leq z \leq x \right\}$$

is expressible as

- (A) $\int_1^2 \int_0^{2/x} \int_0^x dz \, dy \, dx$
- (B) $\int_1^2 \int_0^x \int_0^{2/x} dy \, dz \, dx$
- (C) $\int_0^2 \int_1^z \int_0^{2/x} dy \, dx \, dz$
- (D) $\int_0^2 \int_{\max\{z, 1\}}^2 \int_0^{2/x} dy \, dx \, dz$

Question 33 (JAM 2017 • Q33 • ID: JAM_2017_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. Then which of the following statements is/are TRUE?

- (A) If f is differentiable at $(0,0)$, then all directional derivatives of f exist at $(0,0)$
- (B) If all directional derivatives of f exist at $(0,0)$, then f is differentiable at $(0,0)$
- (C) If all directional derivatives of f exist at $(0,0)$, then f is continuous at $(0,0)$
- (D) If the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous in a disc centered at $(0,0)$, then f is differentiable at $(0,0)$

Question 34 (JAM 2017 • Q34 • ID: JAM_2017_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 If X and Y are $n \times n$ matrices with real entries, then which of the following is/are TRUE?

- (A) If $P^{-1}XP$ is diagonal for some real invertible matrix P , then there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of X
- (B) If X is diagonal with distinct diagonal entries and $XY = YX$, then Y is also diagonal
- (C) If X^2 is diagonal, then X is diagonal
- (D) If X is diagonal and $XY = YX$ for all Y , then $X = \lambda I$ for some $\lambda \in \mathbb{R}$

Question 35 (JAM 2017 • Q35 • ID: JAM_2017_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Let G be a group of order 20 in which the conjugacy classes have sizes 1, 4, 5, 5, 5. Then which of the following is/are TRUE?

- (A) G contains a normal subgroup of order 5
- (B) G contains a non-normal subgroup of order 5
- (C) G contains a subgroup of order 10
- (D) G contains a normal subgroup of order 4

Question 36 (JAM 2017 • Q36 • ID: JAM_2017_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Let $\{x_n\}$ be a real sequence such that $7x_{n+1} = x_n^3 + 6$ for $n \geq 1$. Then which of the following statements is/are TRUE?

- (A) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 1
- (B) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 2
- (C) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to 1
- (D) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to -3

Question 37 (JAM 2017 • Q37 • ID: JAM_2017_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let S be the set of all rational numbers in $(0,1)$. Then which of the following statements is / are TRUE?

- (A) S is a closed subset of $\mathbb{R}$
- (B) S is not a closed subset of $\mathbb{R}$
- (C) S is an open subset of $\mathbb{R}$
- (D) Every $x \in (0,1) \setminus S$ is a limit point of S

Question 38 (JAM 2017 • Q38 • ID: JAM_2017_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let M be an $n \times n$ matrix with real entries such that $M^3 = I$. Suppose that $Mv \neq v$ for any non-zero vector v . Then which of the following statements is / are TRUE?

- (A) M has real eigenvalues
- (B) $M + M^{-1}$ has real eigenvalues
- (C) n is divisible by 2
- (D) n is divisible by 3

Question 39 (JAM 2017 • Q39 • ID: JAM_2017_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = (y-1)(y-3)$$

satisfying the condition $y(0) = 2$. Then which of the following is/are TRUE?

- (A) The function $y(x)$ is not bounded above
- (B) The function $y(x)$ is bounded
- (C) $\lim_{x \rightarrow +\infty} y(x) = 1$
- (D) $\lim_{x \rightarrow -\infty} y(x) = 3$

Question 40 (JAM 2017 • Q40 • ID: JAM_2017_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Let $k, \ell \in \mathbb{R}$ be such that every solution of

$$\frac{d^2 y}{dx^2} + 2k \frac{dy}{dx} + \ell y = 0$$

satisfies $\lim_{x \rightarrow \infty} y(x) = 0$. Then

- (A) $3k^2 + \ell < 0$ and $k > 0$
- (B) $k^2 + \ell > 0$ and $k < 0$
- (C) $k^2 - \ell \leq 0$ and $k > 0$
- (D) $k^2 - \ell > 0, k > 0$ and $\ell > 0$

Question 41 (JAM 2017 • Q41 • ID: JAM_2017_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 If the orthogonal trajectories of the family of ellipses $x^2 + 2y^2 = c_1$, $c_1 > 0$, are given by $y = c_2 x^\alpha$, $c_2 \in \mathbb{R}$, then $\alpha =$ _____

Question 42 (JAM 2017 • Q42 • ID: JAM_2017_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let G be a subgroup of $GL_2(\mathbb{R})$ generated by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$. Then the order of G is _____

Question 43 (JAM 2017 • Q43 • ID: JAM_2017_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Consider the permutations $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 7 & 8 & 6 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 1 & 7 & 6 & 8 & 2 \end{pmatrix}$ in S_8 . The number of $\eta \in S_8$ such that $\eta^{-1}\sigma\eta = \tau$ is equal to _____

Question 44 (JAM 2017 • Q44 • ID: JAM_2017_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let P be the point on the surface $z = \sqrt{x^2 + y^2}$ closest to the point $(4, 2, 0)$. Then the square of the distance between the origin and P is _____

Question 45 (JAM 2017 • Q45 • ID: JAM_2017_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45

$$\left(\int_0^1 x^4 (1-x)^5 dx \right)^{-1} = \underline{\hspace{2cm}}$$

Question 46 (JAM 2017 • Q46 • ID: JAM_2017_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Let M be the matrix whose columns are $v_1, v_2, 2v_1 - v_2, v_1 + 2v_2$ in that order. Then the number of linearly independent solutions of the homogeneous system of linear equations $Mx = 0$ is _____

Question 47 (JAM 2017 • Q47 • ID: JAM_2017_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47

$$\frac{1}{2\pi} \left(\frac{\pi^3}{1!3} - \frac{\pi^5}{3!5} + \frac{\pi^7}{5!7} - \dots + \frac{(-1)^{n-1} \pi^{2n+1}}{(2n-1)!(2n+1)} + \dots \right) = \underline{\hspace{2cm}}$$

Question 48 (JAM 2017 • Q48 • ID: JAM_2017_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let P be a 7×7 matrix of rank 4 with real entries. Let $\mathbf{a} \in \mathbb{R}^7$ be a column vector. Then the rank of $P + \mathbf{a}\mathbf{a}^T$ is at least _____

Question 49 (JAM 2017 • Q49 • ID: JAM_2017_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 For $x > 0$, let $[x]$ denote the greatest integer less than or equal to x . Then

$$\lim_{x \rightarrow 0^+} x \left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \cdots + \left\lfloor \frac{10}{x} \right\rfloor \right) = \text{_____}$$

Question 50 (JAM 2017 • Q50 • ID: JAM_2017_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 The number of subgroups of $\mathbb{Z}_7 \times \mathbb{Z}_7$ of order 7 is _____

Question 51 (JAM 2017 • Q51 • ID: JAM_2017_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51 Let $y(x)$, $x > 0$ be the solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0$$

satisfying the conditions $y(1) = 1$ and $y'(1) = 0$. Then the value of $e^2 y(e)$ is _____

Question 52 (JAM 2017 • Q52 • ID: JAM_2017_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 Let T be the smallest positive real number such that the tangent to the helix

$$\cos t \hat{i} + \sin t \hat{j} + \frac{t}{\sqrt{2}} \hat{k}$$

at $t = T$ is orthogonal to the tangent at $t = 0$. Then the line integral of $\vec{F} = x\hat{j} - y\hat{i}$ along the section of the helix from $t = 0$ to $t = T$ is _____

Question 53 (JAM 2017 • Q53 • ID: JAM_2017_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let $f(x) = \frac{\sin \pi x}{\pi \sin x}$, $x \in (0, \pi)$, and let $x_0 \in (0, \pi)$ be such that $f'(x_0) = 0$. Then

$$(f(x_0))^2 (1 + (\pi^2 - 1) \sin^2 x_0) = \underline{\hspace{2cm}}$$

Question 54 (JAM 2017 • Q54 • ID: JAM_2017_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 The maximum order of a permutation σ in the symmetric group S_{10} is $\underline{\hspace{2cm}}$

Question 55 (JAM 2017 • Q55 • ID: JAM_2017_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55 Let $a_n = \sqrt{n}$, $n \geq 1$, and let $s_n = a_1 + a_2 + \dots + a_n$. Then

$$\lim_{n \rightarrow \infty} \left(\frac{a_n/s_n}{-\ln(1 - a_n/s_n)} \right) = \underline{\hspace{2cm}}$$

Question 56 (JAM 2017 • Q56 • ID: JAM_2017_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 For a real number x , define $[x]$ to be the smallest integer greater than or equal to x . Then

$$\int_0^1 \int_0^1 \int_0^1 ([x] + [y] + [z]) \, dx \, dy \, dz = \underline{\hspace{2cm}}$$

Question 57 (JAM 2017 • Q57 • ID: JAM_2017_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 For $x > 1$, let

$$f(x) = \int_1^x \left(\sqrt{\log t} - \frac{1}{2} \log \sqrt{t} \right) dt$$

The number of tangents to the curve $y = f(x)$ parallel to the line $x + y = 0$ is $\underline{\hspace{2cm}}$

Question 58 (JAM 2017 • Q58 • ID: JAM_2017_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let $\alpha, \beta, \gamma, \delta$ be the eigenvalues of the matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 =$ _____

Question 59 (JAM 2017 • Q59 • ID: JAM_2017_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 The radius of convergence of the power series

$$\sum_{n=0}^{\infty} n! x^{n^2}$$

is _____

Question 60 (JAM 2017 • Q60 • ID: JAM_2017_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If

$$y(x) = \int_{\sqrt{x}}^x \frac{e^t}{t} dt, \quad x > 0$$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2017 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	MCQ	+1	D	21	MCQ	+2	D	41	NAT	+1	1.9 to 2.1
2	MCQ	+1	A	22	MCQ	+2	B	42	NAT	+1	5.9 to 6.1
3	MCQ	+1	B	23	MCQ	+2	B	43	NAT	+1	-0.01 to +0.01
4	MCQ	+1	C	24	MCQ	+2	D	44	NAT	+1	9.9 to 10.1
5	MCQ	+1	B	25	MCQ	+2	C	45	NAT	+1	1259.9 to 1260.1
6	MCQ	+1	D	26	MCQ	+2	A	46	NAT	+1	1.9 to 2.1
7	MCQ	+1	B	27	MCQ	+2	C	47	NAT	+1	0.49 to 0.51
8	MCQ	+1	A	28	MCQ	+2	C	48	NAT	+1	2.9 to 3.1
9	MCQ	+1	A	29	MCQ	+2	C	49	NAT	+1	54.9 to 55.1
10	MCQ	+1	D	30	MCQ	+2	B	50	NAT	+1	7.9 to 8.1
11	MCQ	+2	B	31	MSQ	+2	A, C	51	NAT	+2	2.9 to 3.1
12	MCQ	+2	A	32	MSQ	+2	A, B, D	52	NAT	+2	2.0 to 2.2
13	MCQ	+2	C	33	MSQ	+2	A, D	53	NAT	+2	0.9 to 1.1
14	MCQ	+2	D	34	MSQ	+2	A, B, D	54	NAT	+2	29.9 to 30.1
15	MCQ	+2	C	35	MSQ	+2	A, C	55	NAT	+2	0.9 to 1.1
16	MCQ	+2	A	36	MSQ	+2	A, C	56	NAT	+2	2.9 to 3.1
17	MCQ	+2	A	37	MSQ	+2	B, D	57	NAT	+2	0.9 to 1.1
18	MCQ	+2	D	38	MSQ	+2	B, C	58	NAT	+2	5.9 to 6.1
19	MCQ	+2	D	39	MSQ	+2	B, C, D	59	NAT	+2	0.9 to 1.1
20	MCQ	+2	C	40	MSQ	+2	C, D	60	NAT	+2	1.34 to 1.36

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, -2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.